Priyanshu Pratik

priyanshupratik07@gmail.com | +91 9173592867
ppratik765.github.io | linkedin.com/in/priyanshu-pratik-ai | github.com/Ppratik765

EDUCATION

Gati Shakti Vishwavidyalaya — Vadodara, Gujarat, India
Bachelor of Technology (B.Tech) in Artificial Intelligence and Data Science — Aug 2024 – May 2028
Specialisation: Transportation & Logistics
- **Current Status:** 3rd Year (5th Semester) | CGPA: 8.77
- **Core Coursework:** Deep Learning, Software Engineering for AI Systems (MLOps), Cloud Computing, DBMS, Design & Analysis of Algorithms.
- **Hardware & Architecture:** Studied low-level digital design, circuit synthesis, and hardware architecture utilising MIPS assembly language and Verilog.

Delhi Public School — Vadodara, Gujarat, India
Class XII (Higher Secondary) — Apr 2021 – May 2023
- **Percentage:** 87.0% (435/500)
- **JEE Main:** 96.6 Percentile (AIR: ∼50,000)

Soar Valley College — Leicester, England
Class X (Secondary / GCSE) — Sep 2016 – Jun 2020
- **Academics:** Achieved outstanding academic excellence, securing 7 Grade 9s across my subjects.

TECHNICAL SKILLS

Software & Hardware Engineering: Python, Rust, Kotlin, Java, Javascript, C, SQL, MIPS Assembly, Verilog, Bash/Shell, Linux, REST APIs, Git, Docker, Jetpack Compose, PyQt6, MVVM.

AI & Data Science: PyTorch, TensorFlow Lite, Pandas, NumPy, LangGraph, Agentic Systems, Predictive Analytics, Edge Inference, Time-Series Analysis.

Vision & Signal Processing (DSP): YOLO, SegFormer, ByteTrack, Fast Fourier Transform (FFT), Mel-Spectrograms, Acoustic Fingerprinting, Homography Projection.

Simulation, Systems & Logic: WebGL, Three.js, Vite, FastF1 API, Monte Carlo Engines, Digital Twin Arch, Bayesian Sensor Fusion, Graph Search Algorithms.

EXPERIENCE

BISAG-N [GitHub] — Gandhinagar, Gujarat, India
AI Engineering Intern — Jun 2026 – Jul 2026
- Architected an agentic supply chain simulator, employing a hybrid architecture that integrated deterministic agents with a LangGraph-powered Generative AI advisory board.
- Simulated complex logistical networks to dynamically model and effectively mitigate the bullwhip effect in global supply chain transit.
- Authored comprehensive technical documentation under the direct mentorship of Dr. Manoj Pandya (Additional Director of BISAG-N).

COMPETITIONS & INITIATIVES

LocalPDF Pro (Privacy Utility) [GitHub] — *Dec 2025*
- Constructed a 100% offline, privacy-centric document processing application using PyQt6 and Kotlin, ensuring secure execution without cloud dependency.

Cipher Saga (Cryptic Hunt) — *Mar 2026*
- Competed in a collegiate puzzle competition, analysing and dynamically parsing complex puzzles and level configurations to advance through computational challenges.

PROJECTS

3D ITS Digital Twin [GitHub] — *Jul 2026*
- Developed a modular Digital Twin transforming standard monocular CCTV traffic footage into a real-time, interactive 3D WebGL reconstruction via Three.js.
- Enhanced multi-object tracking and semantic segmentation by integrating YOLO, ByteTrack, SegFormer, and ground-plane homography projection.

Ambulance Corridor [GitHub] — *Jul 2026*
- Implemented an Intelligent Transportation System (ITS) deep learning pipeline utilising Late Bayesian Sensor Fusion for emergency vehicle detection.
- Expanded acoustic siren detection capabilities by combining YOLO visual detection with a PyTorch 2D CNN on Mel-Spectrograms to trigger adaptive traffic preemption.

F1 Monte Carlo Predictor [GitHub] — *Jun 2026*
- Created a CLI Formula 1 race predictor analysing Free Practice and Sprint session data to forecast Grand Prix outcomes.
- Powered by a robust Monte Carlo simulation engine modelling thousands of race scenarios, rendering probabilistic outputs into high-fidelity graphics.

PulmoSense (Edge AI) [GitHub] — *Apr 2026*
- Trained an on-device PyTorch model deployed via TFLite to compute 128-band Mel spectrograms for real-time acoustic edge inference.

Aura (Audio ID) [GitHub] — *Mar 2026*
- Devised an FFT-based acoustic fingerprinting algorithm capable of fault-tolerant pattern recognition and ”Query by Humming” matching.

Waveglider & Vector Squadron [Live 1] [Live 2] — *Feb 2026*
- Programmed interactive WebGL and Rust-based 3D simulations (infinite ocean and starfighter environments) featuring custom physics controllers and optimised low-graphics rendering.

Wayfinder (Pathfinding) [Live] [GitHub] — *Dec 2025*
- Improved route optimization by deploying informed and uninformed search algorithms within a custom visualizer to compute optimal paths across complex map graphs.

Beyond the Apex (Telemetry Platform) [Live] — *Nov 2025*
- Built a platform parsing motorsport telemetry via the FastF1 API to generate standard 2D tracking charts for multi-driver analysis.

PROBLEM SOLVING

- **GATE 2027 Aspirant:** Following a foundational roadmap for the GATE Data Science and Artificial Intelligence (DA) examination.
- **Algorithms:** Solved 160+ data structure problems on LeetCode.

INTERESTS

Deep Learning Architectures, Computer Vision, High-Performance Computing, Distributed Systems, Formula 1, Pattern Recognition, 3D Physics Controllers, Low-Level Hardware Architecture, Space Exploration, Aeronautics.